

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458876.7538 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.04 +/- 1.75 [%]
Orbital Inclination (inc)	89.22 +/- 1.69
Airmass coefficient 1 (a1)	1.15 +/- 0.19
Airmass coefficient 2 (a2)	-0.077 +/- 0.029
Scatter in the residuals of the lightcurve fit is	16.88 %
Best Comparison Star	None
Optimal Aperture	7.53
Optimal Annulus	7.53
Transit Duration (day)	0.104 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.05 vs 0.131 (deviation 0.56 σ)
Duration fit vs published	0.104 d vs 0.1075 d (deviation 0.18 σ)
Mid-time O-C	6.4 min
Depth SNR	2.3
Residual scatter	16.88 %

Light curve

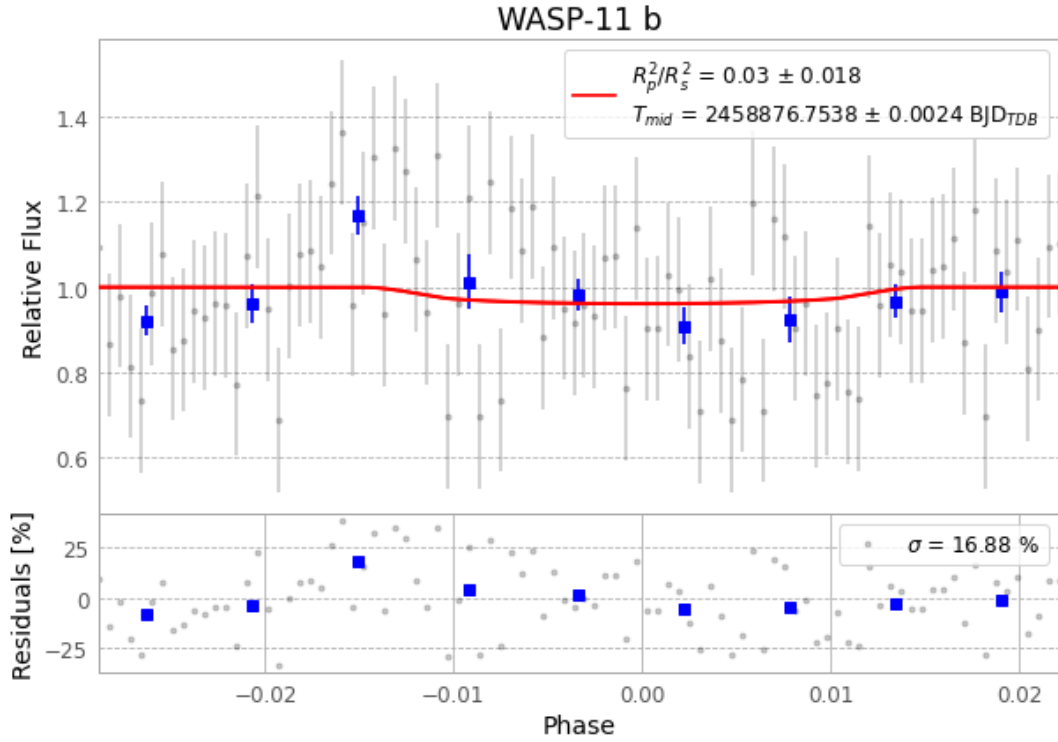


Figure 1 — FinalLightCurve_WASP-11 b_2020-01-27.png (SHA-256 169fc6fab8b194...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [632, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee20390496b16124...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), HATP-10200128032712.FITS → HATP-10200128080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-200128.log (b5881827ead4...), FinalLightCurve_WASP-11 b_2020-01-27.png (169fc6fab8...), FinalLightCurve_WASP-11 b_2020-01-27.csv (6c915d54d737...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 10:12Z.